4-WEEK PROJECT IMPLEMENTATION & ALLOCATION PLAN

Power Peak Minimization for SALBP using Exact SAT & MaxSAT

Course: Modern Issues in Computer Science | Academic Year: 2026 – 2027

1. Problem Overview & Core Objective

SALBP: Assign N tasks to M workstations subject to precedence constraints and cycle time C. **Objective:** Jointly balance the line and schedule start times s_j to minimize peak instantaneous power: min P_max = min max_t sum_{j in active(t)} p_j.

2. 4-Week Master Action Checklist

- **Week 1 (Theory & Data):** Both study papers/SAT; A registers licenses (Gurobi/CPLEX), prepares SALBP dataset & Input/Output spec doc; B sets up environment & LaTeX template (cas-sc).
- **Week 2 (Coding & Theory Paper):** B implements all solvers (PySAT SAT/PB, MaxSAT RC2, Incremental SAT, Gurobi, CP-SAT), runs tests on small instances, hands code to A, and writes Sections 1-3 of the paper.
- **Week 3 (Experiments & Report):** A runs full batch benchmarks on machine & plots cnsplots charts (Cactus plots, Bar charts); B receives CSV + plots, writes Section 4 (Experiments) & Section 5 (Conclusion).
- **Week 4 (Slide, Demo & Defense):** B designs 12-15 slide presentation (inserting A's plots) & prepares live demo; both rehearse defense (A: Data/Plots/Results, B: Theory/Demo) and submit.

3. 4-Step Handover Workflow (Member A & Member B)

Week / Step	Actor	Key Actions & Deliverables
Week 1	Both + A Lead	Both study problem & paper. A registers Gurobi/CPLEX academic keys, prepares dataset & writes Input/Output spec doc for B.
Week 2	Member B	B implements all solvers (PySAT SAT/PB, MaxSAT, Incremental, Gurobi, CP-SAT), tests on small instances, hands code to A, and drafts Sections 1,2,3.
Week 3	A (Run+Plot) + B (Write)	A runs batch benchmarks on machine, plots cnsplots charts & sends CSV + figures to B. B completes Sections 4 & 5.
Week 4	B (Slide) + Both	B builds presentation slides (using A's plots) & live demo. Both review paper PDF, rehearse defense (A: Data/Plots/Results, B: Theory/Demo) & submit.

4. Detailed Responsibilities Allocation Table

MEMBER A: Data, License, Runner & Plotting Lead	MEMBER B: Lead Coder, Paper Writer & Slide/Demo Lead
• **W1:** Study papers; register Gurobi/CPLEX; prepare dataset; write Input/Output spec doc. • **W2:** Review data format, prepare workstation for batch runs. • **W3:** Execute batch benchmarks on machine; **plot cnsplots charts** & send CSV + figures to B. • **W4:** Review paper results; present Introduction, Dataset, Plots & Results.	• **W1:** Study papers; setup environment & LaTeX template. • **W2:** Implement SAT/PB, MaxSAT, Incremental, Gurobi, CP-SAT; test on small instances & hand to A; write Sections 1-3. • **W3:** Insert A's plots into paper and write Sections 4 & 5. • **W4:** Build presentation slides (with A's plots); prepare Live Demo; present Theory & Demo.

5. 4-Week Detailed Timeline

Week / Phase	Member A Activities	Member B Activities
Week 1 (Theory & Prep)	Study paper; register licenses; prepare dataset & Input/Output spec.	Study paper; setup environment & LaTeX cas-sc template.
Week 2 (Coding & Theory)	Review data structures, prepare workstation for runs.	Implement all solvers; test & hand code to A; write Sections 1, 2, 3.
Week 3 (Run, Plot & Report)	Run batch benchmarks on machine; **plot cnsplots charts**; export CSV + figures to B.	Insert figures; write Section 4 (Experiments & Analysis) & Section 5 (Conclusion).
Week 4 (Slide & Defense)	Review final paper; rehearse defense (Data, Plots & Results).	Build slide deck (with A's plots); prepare Live Demo; rehearse defense (Theory/Demo) & submit.